

CS 463G & CS 663: Introduction to AI

Midterm Exam

October 12, 2026

Please print your name and UKID: _____

Important Information:

- The duration of this exam is 50 minutes.
- This exam accounts for 20% of the total course score, equivalent to 20 points.
- There are 24 multiple-choice questions. Each correct answer earns 1 point. If you answer 20 or more questions correctly, you will earn the full 20 points for this exam.
- No books, notes, calculators, or cheat sheets are allowed.
- Do not distribute or disclose the contents of this exam.

Darken the bubbles with a pencil. If you make a mistake, be sure to erase the mark completely.
Select one answer per question.



The **LAST** four digits of your UKID, in order from top row to bottom row:

0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9

1. (A) (B) (C) (D)	9. (A) (B) (C) (D)	17. (A) (B) (C) (D)
2. (A) (B) (C) (D)	10. (A) (B) (C) (D)	18. (A) (B) (C) (D)
3. (A) (B) (C) (D)	11. (A) (B) (C) (D)	19. (A) (B) (C) (D)
4. (A) (B) (C) (D)	12. (A) (B) (C) (D)	20. (A) (B) (C) (D)
5. (A) (B) (C) (D)	13. (A) (B) (C) (D)	21. (A) (B) (C) (D)
6. (A) (B) (C) (D)	14. (A) (B) (C) (D)	22. (A) (B) (C) (D)
7. (A) (B) (C) (D)	15. (A) (B) (C) (D)	23. (A) (B) (C) (D)
8. (A) (B) (C) (D)	16. (A) (B) (C) (D)	24. (A) (B) (C) (D)



Midterm Preparation Guide

Question-by-question topics

Page 1 reproduces the exam's front page. Scope: Lectures 1–16, including Lecture 2b. Each entry matches its exam question number. Practice with lecture notes, code demonstrations, and the released previous exam.

1. **AI approaches and cognitive modeling** (Lec. 1). Identify a research objective that models human cognition rather than external behavior alone.
2. **PEAS and performance measures** (Lec. 2–2b). Distinguish outcome criteria from sensors, actuators, and environment.
3. **Rational decisions and expected performance** (Lec. 2–2b). Compute and compare simple expected rewards. The calculation rule will be supplied.
4. **Model-based agent architecture** (Lec. 2b). Explain the role of memory and a world model with limited sensing.
5. **Environment properties** (Lec. 3). Classify observability, whether actions affect later decisions, and change during deliberation.
6. **Sufficient search-state representation** (Lec. 4). Retain the information that affects future legal actions.
7. **Breadth-first search trace** (Lec. 5). Trace a FIFO frontier with a stated child order.
8. **Depth-first search and backtracking trace** (Lec. 5). Trace recursive visits and backtracking with a stated child order.
9. **Uniform-cost search and goal-test timing** (Lec. 6). Track cheaper routes and apply the goal test at the stated time.
10. **A* frontier priorities** (Lec. 7–8). Compare frontier entries using path costs and estimates.
11. **Admissibility and consistency** (Lec. 7–8). Check overestimation and a local consistency inequality.
12. **Repeated states and improved paths in A*** (Lec. 7–8). Update best path costs and handle outdated frontier entries.
13. **Minimax backup and move selection** (Lec. 9). Back up MIN/MAX utilities and select the root move.
14. **Alpha-beta cutoff trace** (Lec. 10). Carry bounds and identify a cutoff in a stated evaluation order.
15. **Hill climbing and local maxima** (Lec. 11). Distinguish a local stopping point from a global optimum.
16. **Simulated annealing and temperature** (Lec. 11). Compare acceptance of the same worse move at different temperatures.
17. **One-point genetic crossover** (Lec. 12). Construct an offspring from the specified parent segments.
18. **CSP domains and constraint checking** (Lec. 13). Test assignments against domains, unary rules, and binary constraints.
19. **Directed arc revision and support** (Lec. 14). Remove unsupported values in the specified arc direction.
20. **MRV with the degree tie-breaker** (Lec. 15). Compare current domain sizes, then unassigned-neighbor counts.
21. **Least-constraining value by deletion counts** (Lec. 15). Compare the neighbor values deleted by forward checking.
22. **Forward checking and its propagation limit** (Lec. 15). Trace one-step neighbor-domain deletions and explain which constraints forward checking does not propagate.
23. **Implication and counterexamples** (Lec. 16). Find a truth assignment that violates an implication.
24. **Negation scope and De Morgan's law** (Lec. 16). Distinguish not-both from neither and apply De Morgan's law.